MarginMix

Decision Engine — Scoring & Rules Logic

Complete reference for the deterministic margin risk assessment system

Generated: November 1, 2026

This reference documents the standalone full assessment's deterministic Model1 computation. The scorecard is authoritative for question points, signal groups, arithmetic means, weights and verdict thresholds. The updated 55 example is a regression fixture, not a target for other answers. AI does not calculate or supply scores.

LAYER 1 — QUESTIONS TO SIGNALS

The scorecard assigns points to 18 answers. Q1–Q6 are identifiers/context and are not in the composite. Q23 is a confidence override, not a numeric dimension input. Q27 is narrative-only. The full assessment is independent of the homepage quick profiler; answers are not prefilled or skipped.

Context Questions — Not in the Composite

Q1 Full Name · Q2 Work Email · Q3 Role / Title · Q4 Organisation Name · Q5 Organisation Size · Q6 Industry · Q8 Context · conditional Q10b Delivery Model

Model1 Scored Answers

Q#	Answer	Points
Q7	Decision: new / renewal / exploratory; expansion / escalation; strategic exception	0 / 100 / 50
Q9	50: new / under 6 months / renewal-expansion; 0: 6–12 months / 12+ months	50 / 0
Q10	Retainer; hybrid / T&M / managed; fixed fee / commission / outcome-based	0 / 50 / 100
Q11, Q12, Q14, Q16, Q17, Q18	Low / medium / high	0 / 50 / 100
Q13	Minimal / periodic / frequent / continuous senior involvement	0 / 50 / 100 / 100
Q15	Execution / balanced / thinking; advisory or strategy; technical implementation	0 / 50 / 100; 100; 50
Q19	Junior execution / mid-level production / senior thinking / no clear path	0 / 50 / 100 / 100
Q20	Significant / some / minimal / no meaningful additional value	0 / 50 / 100 / 100
Q21	Less than usual / about the same / more than usual senior oversight	0 / 50 / 100
Q22	Minimal / moderate / heavy coordination drag	0 / 50 / 100
Q24	Agentic framework: no / yes / NA	0 / 100 / 0
Q25	AI hours replaced: 0–25% / 25–50% / 50–75% / 75–100% / NA	0 / 50 / 100 / 100 / 0
Q26	Commercial AI impact measured: yes / no / NA	0 / 100 / 0

Q23	Low confidence triggers the override; Q27 open text is narrative-only.	Override / no points

Scorecard interpretation

Points are risk points, not dimension bands. Hybrid pricing is one answer worth 50 points; it is not separately decomposed into components. Fixed fee scores 100 across industries. Optional missing AI follow-ups use the existing NA mapping (0); unsupported answers fail explicitly rather than receiving invented points.

LAYER 2 — SIGNALS TO NUMERIC DIMENSIONS

Each named signal is first scored from its mapped answer. Signals joined by MAX use the highest point value. Each dimension is the arithmetic mean of its listed signal scores; no categorical dimension conversion is used for scoring.

Dimension	Signal and source question	Aggregation
Workforce Intensity	SeniorDependency = MAX(Q13,Q21) IterationLoad proxy = Q15 IterationLoad = Q16 GovernanceStrength = Q20	Mean of 4 signal scores
Coordination Entropy	CoordinationIntensity = Q14 CoordinationLoad = MAX(Q18,Q22)	Mean of 2 signal scores
Commercial Exposure	EngagementState = MAX(Q7,Q9) PricingRigidity = Q10 ScopeElasticity = Q17	Mean of 3 signal scores
Volatility Control	ClientVolatility = Q11 StakeholderLoad = Q12	Mean of 2 signal scores
AI Substitution and Measurement Risk	AILeverage = Q19 AIRiskPeak = MAX(Q24,Q25,Q26)	Mean of 2 signal scores

LAYER 3 — DIMENSIONS TO VERDICT

The raw composite is the sum of each numeric dimension mean multiplied by its unchanged weight. Preserve full precision through verdict selection; round only the displayed composite to one decimal place. The updated sample 55 is a fixture, not a score to impose on other answer sets. Its Q9 answer is ongoing under six months (50 points); six to twelve months remains 0.

Step 1 — Hard Override (fires before scoring)

Override	Condition	Verdict	Trigger Code
Confidence Failure	Fires when the existing Confidence Signal is Negative: low Q23 delivery confidence, or minimal/no additional value on Q20. This override is evaluated before the weighted verdict.	Do Not Proceed Without Repricing	LOW_CONFIDENCE_OVERRIDE

Step 2 — Weighted Score (applied when no override fires)

Dimension means are already numeric; do not convert them to bands and back into points. For each dimension, weighted contribution = dimension mean × weight ÷ 100. Composite = sum of the five contributions. The override does not replace or fabricate the underlying composite.

Dimension	Weight	Score formula	Contribution
Workforce Intensity	30%	Mean(Q13/Q21 MAX, Q15, Q16, Q20)	Mean × 0.30
Coordination Entropy	25%	Mean(Q14, Q18/Q22 MAX)	Mean × 0.25

Commercial Exposure	20%	Mean(Q7/Q9 MAX, Q10, Q17)	Mean × 0.20
Volatility Control	15%	Mean(Q11, Q12)	Mean × 0.15
AI Substitution and Measurement Risk	10%	Mean(Q19, Q24/Q25/Q26 MAX)	Mean × 0.10
TOTAL	100%	Sum of five contributions	Composite 0–100

Illustrative regression fixture — not a target

Dimension	Mean	Weight	Contribution
Workforce Intensity	62.5	30%	18.75
Coordination Entropy	50	25%	12.5
Commercial Exposure	50	20%	10
Volatility Control	25	15%	3.75
AI Substitution and Measurement Risk	100	10%	10
Raw total			55

Those values come from one fixed answer fixture. Other answers must produce their own signal scores, means and weighted contributions.

Score to Verdict Thresholds

Score Range	Verdict	Trigger Code
0 to <20	Structurally Safe	WEIGHTED_SCORE_SAFE
20 to <40	Execution Heavy	WEIGHTED_SCORE_EXECUTION_HEAVY
40 to <60	Price Sensitive	WEIGHTED_SCORE_PRICE_SENSITIVE
60 to <80	Structurally Fragile	WEIGHTED_SCORE_FRAGILE
80 to 100	Do Not Proceed Without Repricing	WEIGHTED_SCORE_CRITICAL

Verdict Meanings

- **Structurally Safe:** Engagement viable under stated assumptions. Proceed with standard governance and monitoring.
- **Execution Heavy:** Senior dependency and iteration load will dominate effort. Elevated management attention required throughout delivery.
- **Price Sensitive:** High scope elasticity or pricing rigidity creates commercial vulnerability. Pricing assumptions require explicit protection.
- **Structurally Fragile:** High workforce intensity combined with high coordination entropy creates unsustainable margin pressure. Structural changes required.
- **Do Not Proceed Without Repricing:** Pricing or delivery confidence is insufficient to proceed safely. Engagement requires repricing or structural changes before commitment.

The confidence override takes priority over every weighted-score band. There is no separate first-match structural-overload verdict rule.

CORE DESIGN PRINCIPLES

- **Deterministic scoring:** Every point, signal maximum, dimension mean and weighted contribution is calculated from the submitted answers. AI is limited to prose; numeric claims it produces are rejected.
- **Confidence override:** The existing negative Confidence Signal remains a pre-scoring override. It changes the verdict, not the numeric composite.
- **No margin forecast:** The assessment does not estimate margin erosion or effective margin, and does not invent staffing percentages or effort allocations.
- **Separate quick profiler:** The full assessment at /assessment has its own independent question, scoring and report flow. The homepage profiler does not prefill it or skip its questions.